

Select Problems in Stochastic Calculus

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1 Prove that every stationary Gaussian Markov process is an Ornstein-Uhlenbeck process.

We will approach this problem by using a result from Basu's textbook:

Lemma: A zero-mean Gaussian process has the Markov property iff its covariance satisfies $C(s, u)C(t, t) = C(s, t)C(t, u)$ when $s < t < u$.

Proof:

Forward direction: Assume $X_t \sim \mathcal{N}(0, C)$ is Markov with respect to the filtration $\mathcal{F}_t := \sigma(X_s, s \leq t)$. For Gaussian random variables X, Y ,

$$\mathbb{E}[X|Y] = \mathbb{E}[X] + \Sigma_{XY}\Sigma_Y^{-1}(Y - \mathbb{E}[Y])$$

where Σ is the covariance matrix between X, Y . So we get

$$\mathbb{E}[X_u|\mathcal{F}_s] = \mathbb{E}[X_u|X_s] = \frac{C(s, u)}{C(s, s)}X_s$$

But also, by the tower law,

$$\begin{aligned}\mathbb{E}[X_u|\mathcal{F}_s] &= \mathbb{E}[\mathbb{E}[X_u|\mathcal{F}_t]|\mathcal{F}_s] \\ &= \mathbb{E}\left[\frac{C(t, u)}{C(t, t)}X_t|\mathcal{F}_s\right] \\ &= \frac{C(t, u)C(s, t)}{C(t, t)C(s, s)}X_s\end{aligned}$$

yielding $C(s, u)C(t, t) = C(s, t)C(t, u)$ as desired.

Backward direction: Assume $C(s, u)C(t, t) = C(s, t)C(t, u)$ holds $\forall s < t < u$. By the nature of conditional expectation, $X_u - \mathbb{E}[X_u|X_t] \perp X_t$, so

$$X_u - \frac{C(t, u)}{C(t, t)}X_t \perp X_t$$

Computing the covariance between X_s and $X_u - \frac{C(t, u)}{C(t, t)}X_t$ for $s < t$,

$$\mathbb{E}\left[\left(X_u - \frac{C(t, u)}{C(t, t)}X_t\right)X_s\right] = C(s, u) - \frac{C(t, u)C(s, t)}{C(t, t)} = 0$$

The Gaussians are uncorrelated so $X_u - \mathbb{E}[X_u|X_t] \perp X_s \forall s < t$. It remains to show $\mathbb{E}[f(X_u)|\mathbb{F}_t] = \mathbb{E}[f(X_u)|X_t]$ which is equivalent to the Markov property if it holds for all bounded, measurable f .

$$\begin{aligned}\mathbb{E}[f(X_u)|\mathcal{F}_t] &= \mathbb{E}[f(\mathbb{E}[X_u|X_t] + X_u - \mathbb{E}[X_u|X_t])|\mathcal{F}_t] \\ &= \mathbb{E}[f(Z + y)]_{y=\mathbb{E}[X_u|X_t]}\end{aligned}$$

by letting $Y := \mathbb{E}[X_u|X_t]$, $Z := X_u - \mathbb{E}[X_u|X_t]$. Note that Y is measurable with respect to $\mathcal{F}_t$ and Z is independent of it, so by the substitution lemma, $\mathbb{E}[f(Z + Y)]$ conditions only on X_t which holds $\forall t \leq u$, which gives X_t its Markov property.

To be specific, $Z \perp \mathcal{F}_t$ arises from $\text{Cov}(X_u - \mathbb{E}[X_u|X_t], X_s) = 0 \ \forall s \leq t$ so Z is independent of all linear combinations of X_s as well, and $\mathcal{F}_t$ is spanned by the closure of linear combinations of Gaussians X_s . $\square$

Back to the actual problem, an Ornstein-Uhlenbeck process is a stationary process with mean μ and covariance $C(s, t) = \frac{\sigma^2}{2\theta} e^{-\theta|t-s|}$ governed by the SDE

$$dX_t = \theta(\mu - X_t)dt + \sigma dW_t, \theta > 0$$

Suppose X_t is Gaussian, stationary and Markov. By the lemma, $C(s, u)C(t, t) = C(s, t)C(t, u)$, and by stationarity, $C(|u - s|)C(0) = C(|t - s|)C(|u - t|)$ for $s < t < u$. Let $x := |t - s|$, $y := |u - t|$, $h(x) := \frac{C(x)}{C(0)}$. Then,

$$C(x + y)C(0) = C(x)C(y)$$

and dividing by $C(0)^2$ on each side,

$$h(x + y) = h(x)h(y)$$

Since h is an adaptation of a Gaussian covariance function, it is continuous and satisfies Cauchy's exponential functional equation, which has solutions $h(x) = 0$ or $h(x) = e^{cx}$. Observe that we get the equations

$$\begin{aligned} h(x) &= h(1 + 1 + \dots + 1) = h(1)^x & x \in \mathbb{Z} \\ h\left(\frac{p}{q}\right) &= h\left(\frac{1}{q} + \dots + \frac{1}{q}\right) = h\left(\frac{1}{q}\right)^p & x \in \mathbb{Q} \end{aligned}$$

Since $h(1) = h(\frac{1}{q})^q$, $h(\frac{p}{q}) = h(1)^{\frac{p}{q}}$. By density of $\mathbb{Q}$ and continuity, $h(x) = h(1)^x = e^{x \ln h(1)} \ \forall x > 0$, so

$$C(x) = C(0)e^{x \ln\left(\frac{C(1)}{C(0)}\right)}$$

Recall that we have defined x as a distance between times due to stationarity of the process, so $C(x) = C(-x) = C(|t - s|)$ for $s, t \in \mathbb{R}$. The equation becomes

$$C(x) = C(0)e^{|x| \ln\left(\frac{C(1)}{C(0)}\right)}$$

Let $t = s + 1$, so $C(1) = C(s, t)$, $C(0) = C(s, s)$. Since $|\rho| := \frac{|C(s, t)|}{\sqrt{C(s, s)C(t, t)}} \leq 1$, $C(1) \leq C(0)$ so $\ln\left(\frac{C(1)}{C(0)}\right) \leq 0$.

Define $\theta := -\ln\left(\frac{C(1)}{C(0)}\right)$, $\sigma^2 := 2\theta C(0)$ to give

$$C(s, t) = C(|t - s|) = \frac{\sigma^2}{2\theta} e^{-\theta|t-s|}$$

which is precisely the covariance function of the O-U process.

To deal with the mean μ , since the process is stationary, μ is constant for all t , so we can consider the zero-mean $Y_t = X_t - \mu$, which is a O-U process. Then, adding the constant mean back just yields a shifted O-U SDE, as the covariance structure doesn't change if we subtract a mean. Since these are Gaussians, they are uniquely determined by μ, C . Because the covariances match, effectively all stationary Gaussian Markov processes are O-U processes. $\square$

2 Compute the distribution of

$$\int_0^\tau f(s)dB(s)$$

under the conditions: $B(t)$ is a Brownian motion, f is progressively measurable, $\mathbb{E}[\int_0^t f^2(s)ds] < \infty$, $\int_0^\infty f^2(s)ds = \infty$ almost surely, and the stopping time is defined as:

$$\tau := \inf \left\{ t \geq 0 : \int_0^t f^2(s)ds = \sigma^2 \right\}$$

This problem can be solved using a relatively direct application of the Dubins-Schwarz theorem, but first we need to set up some things.

Consider the martingale $M_t = \int_0^t f(s)dB_s$; we will show that its quadratic variation, defined as $\lim_{|\Pi| \downarrow 0} \sum_{i=1}^n (M_{t_i} - M_{t_{i-1}})^2$ is $\int_0^t f^2 ds$ via convergence in probability.

$$\begin{aligned} \mathbb{P} \left(\left| \sum_{i=1}^n (M_{t_i} - M_{t_{i-1}})^2 - \int_0^t f^2(s)ds \right| > \epsilon \right) &= \mathbb{P} \left(\left| \sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f(s)dB_s \right)^2 - \int_0^t f^2(s)ds \right| > \epsilon \right) \\ &\leq \frac{1}{\epsilon^2} \text{Var} \left(\sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f(s)dB_s \right)^2 \right) \\ &= \frac{1}{\epsilon^2} \sum_{i=1}^n \left(\text{Var} \left(\int_{t_{i-1}}^{t_i} f(s)dB_s \right)^2 \right) \\ &= \frac{1}{\epsilon^2} \sum_{i=1}^n \left(3 \text{Var} \left(\int f dB \right)^2 - \text{Var} \left(\int f dB \right)^2 \right) \\ &= \frac{2}{\epsilon^2} \sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f^2(s)ds \right)^2 \end{aligned}$$

where the third step is justified by independence of disjoint Brownian increments, the fourth step is justified by Isserlis-Wick identity, and the fifth step follows from Ito's isometry.

Since $f \in L^2$ on finite time domains,

$$\sum_{i=1}^n \left(\int_{t_{i-1}}^{t_i} f^2(s)ds \right)^2 \leq \sup_n \left[\int_{t_{i-1}}^{t_i} f^2(s)ds \right] \int_0^t f^2(s)ds$$

with the square integral being uniformly continuous on a compact set and thus converging to 0 as $n \rightarrow \infty$. Thus, the quadratic variation of M_t at infinity, $QV(M_t)_\infty = \infty$ by assumption.

Since $\int_0^t f^2 ds$ is continuous and non-decreasing, the stopping time τ is equivalent to

$$\tau = \inf \left\{ t \geq 0 : \int_0^t f^2(s, \omega)ds > \sigma^2 \right\}$$

Now we are ready to make use of the theorem.

Lemma (Dubins-Schwarz): Let M_t be a $\mathcal{F}_t$ -adapted continuous local martingale (with respect to the usual filtration), and its quadratic variation $QV(M)_\infty = \infty$, $M_0 = 0$. Define $\forall t \geq 0$, $\tau_t := \inf\{s : QV(M)_s > t\}$. Then M_{τ_t} is a $\mathcal{F}_{\tau_t}$ -adapted Brownian motion, with $M_t = B_{QV(M)_t}$.

Proof:

We seek to show that M_{τ_t} is a continuous martingale with quadratic variation equal to t , using the Levy characterization of Brownian motion.

Let $W_s := M_{\tau_s}$ be adapted to $\mathcal{F}_{\tau_s}$, the filtration of stopping times $\tau_s := \inf \left\{ t \geq 0 : \int_0^t f^2 > s \right\}$. Clearly, $s \mapsto \tau_s$ is continuous and non-decreasing. Observe that if $\int_0^t f^2 du$ is constant on $[t_1, t_2]$, then $QV(M)_{t_2} - QV(M)_{t_1} = 0$. Since $\int_0^t f^2 du$ is constant, $\int_0^t f dB_u$ is constant almost surely, which is a characteristic of martingales—thus $M_{t_1} = M_{t_2}$ almost surely. Also, M_{τ_s} is continuous because it is a composition of M, τ , both continuous functions.

We also know that M_{τ_s} is a martingale on $\mathcal{F}_{\tau_s}$ by optional stopping. Furthermore,

$$QV(M)_{\tau_s} = \int_0^{\tau_s} f^2 du = s$$

by definition, so M_{τ_s} is a Brownian motion B_s . To conclude the final result,

$$B_{QV(M)_t} = M_{\tau_{QV(M)_t}}$$

almost surely $\forall t$, and $\tau_{QV(M)_t} = t$ because intuitively, τ_s and $QV(M)_t$ are continuous and thus takes the same value "near" the stopping time. $\square$

Using the theorem, $M_\tau = \int_0^\tau f dB_s$ is a Brownian motion, and it is easy to see from definitions that

$$\int_0^\tau f dB_s = M_\tau = B_{QV(M)_\tau} = B_{\int_0^\tau f^2 ds} = B_{\sigma^2}$$

Thus,

$$\int_0^\tau f(s, \omega) dB_s \sim \mathcal{N}(0, \sigma^2)$$

- 3 Let $B(t) = (B_1(t), B_2(t))$ be the two-dimensional Brownian bridge on $[0,1]$ from $(0,0)$ to $(0,0)$, i.e. B_1, B_2 are two independent Brownian bridges on $[0,1]$ —the zero-mean Gaussian process with covariance $\mathbb{E}[B(t)B(s)] = s \wedge t - st$. The modified Levy's stochastic area is defined as:**

$$A := \frac{1}{2} \int_0^1 B_1 dB_2 - B_2 dB_1 = \int_0^1 B_1 dB_2 = \lim_{n \rightarrow \infty} A_n$$

$$A_n = \sum_{k=0}^{n-1} B_1\left(\frac{k}{n}\right) \left(B_2\left(\frac{k+1}{n}\right) - B_2\left(\frac{k}{n}\right) \right)$$

1. Prove the limit characterization in the identity

2. Compute the characteristic function of the Levy area

Part 1: We seek to show that

$$\lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} B_1\left(\frac{k}{n}\right) \left(B_2\left(\frac{k+1}{n}\right) - B_2\left(\frac{k}{n}\right) \right) = \int_0^1 B_1 dB_2$$

where

$$B_1(t) = W_1(t) - tW_1(1)$$

$$B_2(t) = W_2(t) - tW_2(1)$$

with W_1, W_2 independent Brownian motions. Expand the identity:

$$\begin{aligned}
& \lim_{n \rightarrow \infty} \sum_{k=0}^{n-1} \left(W_1 \left(\frac{k}{n} \right) - \frac{k}{n} W_1(1) \right) \left(W_2 \left(\frac{k+1}{n} \right) - \frac{k+1}{n} W_2(1) - W_2 \left(\frac{k}{n} \right) + \frac{k}{n} W_2(1) \right) \\
&= \lim_{n \rightarrow \infty} \underbrace{\sum_{k=0}^{n-1} W_1 \left(\frac{k}{n} \right) \left(W_2 \left(\frac{k+1}{n} \right) - W_2 \left(\frac{k}{n} \right) \right)}_{(I)} - \underbrace{\frac{W_2(1)}{n} \sum_{k=0}^{n-1} W_1 \left(\frac{k}{n} \right)}_{(II)} \\
&\quad - \underbrace{\frac{W_1(1)}{n} \sum_{k=0}^{n-1} k \left(W_2 \left(\frac{k+1}{n} \right) - W_2 \left(\frac{k}{n} \right) \right)}_{(III)} + \underbrace{\frac{W_1(1)W_2(1)}{n^2} \sum_{k=0}^{n-1} k}_{(IV)}
\end{aligned}$$

As $n \rightarrow \infty$,

$$\begin{aligned}
(I) &\rightarrow \int_0^1 W_1(s) dW_2(s) \\
(II) &\rightarrow -W_2(1) \int_0^1 W_1(s) ds \\
(III) &\rightarrow -W_1(1) \int_0^1 s dW_2(s) \\
(IV) &\rightarrow \frac{W_1(1)W_2(1)}{2}
\end{aligned}$$

Applying integration by parts to (III),

$$\lim_{n \rightarrow \infty} A_n = \int_0^1 W_1(s) dW_2(s) - W_2(1) \int_0^1 W_1(s) ds - W_1(1) \left[W_2(1) - \int_0^1 W_2(s) ds \right] + \frac{W_1(1)W_2(1)}{2}$$

Ito's lemma on B_2 gives us $dB_2(s) = dW_2(s) - W_2(1)ds$, so

$$\begin{aligned}
\int_0^1 B_1 dB_2 &= \int_0^1 W_1(s) dB_2(s) - \int_0^1 s W_1(1) dB_2(s) \\
&= \int_0^1 W_1(s) dW_2(s) - W_2(1) \int_0^1 W_1(s) ds - W_1(1) \int_0^1 s dW_2(s) - W_1(1)W_2(1) \underbrace{\int_0^1 s ds}_{=\frac{1}{2}}
\end{aligned}$$

Since the two equations above are equal, $\lim_{n \rightarrow \infty} A_n = \int_0^1 B_1 dB_2$ as desired. $\square$

Part 2: We seek to compute $\mathbb{E}[e^{i\lambda A}]$, which we will first do by evaluating $\mathbb{E}[e^{i\lambda A_n}] = \mathbb{E}[\mathbb{E}[e^{i\lambda A_n} | B_1]]$ and justifying taking a limit. Consider $A_n | B_1$, where the path of B_1 is deterministic:

$$A_n | B_1 = \sum_{k=0}^{n-1} \underbrace{c_k}_{B_1\left(\frac{k}{n}\right)} \left(B_2 \left(\frac{k+1}{n} \right) - B_2 \left(\frac{k}{n} \right) \right)$$

Clearly, $\mathbb{E}[A_n | B_1] = 0$, so this is a zero-mean Gaussian. To evaluate its variance, we need the covariance structure of Brownian bridge increments: $\mathbb{E}[B_2(s)B_2(t)] = s \wedge t - st$. Let $\Delta B_2^k := B_2 \left(\frac{k+1}{n} \right) - B_2 \left(\frac{k}{n} \right)$. Then,

$$\mathbb{E}[\Delta B_2^k \Delta B_2^j] = \mathbb{E} \left[B_2 \left(\frac{k+1}{n} \right) B_2 \left(\frac{j+1}{n} \right) - B_2 \left(\frac{k}{n} \right) B_2 \left(\frac{j+1}{n} \right) - B_2 \left(\frac{j}{n} \right) B_2 \left(\frac{k+1}{n} \right) + B_2 \left(\frac{k}{n} \right) B_2 \left(\frac{j}{n} \right) \right]$$

Proceed by exhausting all integer cases of j, k . Assume $j < k$, so $j \wedge k = j$, $(j+1) \wedge k = j+1$. WLOG, the $k < j$ case is equivalent. Then assume $j = k$ in which case the covariance becomes $\text{Var}(\Delta B_2^k)$. Computing

via the covariance structure gives us

$$\mathbb{E}[\Delta B_2^k \Delta B_2^j] = \begin{cases} -\frac{1}{n^2} & j \neq k \\ \frac{1}{n} - \frac{1}{n^2} & j = k \end{cases}$$

Thus, the increments $B_2\left(\frac{k+1}{n}\right) - B_2\left(\frac{k}{n}\right) \sim \mathcal{N}(0, \Sigma)$, where $\Sigma_{ij} = \mathbb{E}[\Delta B_2^i \Delta B_2^j]$.

So $A_n | B_1 \sim \mathcal{N}(0, \sigma^2)$ as it is a sum of Gaussians, where

$$\begin{aligned} \sigma^2 &= \sum_{k=0}^{n-1} c_k^2 \text{Var}(\Delta B_2^k) + \sum_{k \neq j}^{n-1} c_k c_j \mathbb{E}[\Delta B_2^k \Delta B_2^j] \\ &= \left(\frac{1}{n} - \frac{1}{n^2}\right) \sum_k c_k^2 - \frac{1}{n^2} \sum_{k \neq j} c_k c_j \\ &= \left(\frac{1}{n} - \frac{1}{n^2}\right) \sum_k c_k^2 - \frac{1}{n^2} \left(\left(\sum_k c_k \right)^2 - \sum_k c_k^2 \right) \\ &= \frac{1}{n} \sum_k c_k^2 - \frac{1}{n^2} \left(\sum_k c_k \right)^2 \end{aligned}$$

Also, the characteristic function of a zero-mean gaussian is given by $\phi(\lambda) = \mathbb{E}[e^{i\lambda x}] = e^{-\frac{\sigma^2 \lambda^2}{2}}$, so

$$\mathbb{E}[e^{i\lambda A_n} | B_1] = e^{-\frac{\lambda^2}{2} \left(\frac{1}{n} \sum_k B_1\left(\frac{k}{n}\right)^2 - \frac{1}{n^2} \left(\sum_k B_1\left(\frac{k}{n}\right) \right)^2 \right)}$$

Now we must take

$$\lim_{n \rightarrow \infty} \mathbb{E} \left[e^{-\frac{\lambda^2}{2} \left(\frac{1}{n} \sum_k B_1\left(\frac{k}{n}\right)^2 - \frac{1}{n^2} \left(\sum_k B_1\left(\frac{k}{n}\right) \right)^2 \right)} \right]$$

By bounded convergence theorem on the exponential, we can swap the expectation and limit. As $n \rightarrow \infty$,

$$\begin{aligned} \frac{1}{n} \sum_k B_1^2\left(\frac{k}{n}\right) &\rightarrow \int_0^1 B_1^2(s) ds \\ -\frac{1}{n^2} \left(\sum_k B_1\left(\frac{k}{n}\right) \right)^2 &\rightarrow - \left(\int_0^1 B_1(s) ds \right)^2 \end{aligned}$$

And we get

$$\mathbb{E}[e^{i\lambda A}] = \mathbb{E} \left[e^{-\frac{\lambda^2}{2} \left(\int_0^1 B_1^2(s) ds - \left(\int_0^1 B_1(s) ds \right)^2 \right)} \right]$$

Define $W := \int_0^1 B_1^2 ds - \left(\int_0^1 B_1 ds \right)^2$. To simplify this expression, we appeal to the Karhunen-Loeve expansion of the Brownian bridge,

$$B_1(s) = \sum_{k=1}^{\infty} \frac{\sqrt{2}}{\pi k} \sin(\pi k s) \xi_k$$

where $\xi_k \sim \mathcal{N}(0, 1)$ i.i.d. across k . Recall that $\sqrt{2} \sin(\pi k s)$ are orthonormal in $L^2([0, 1])$ with respect to the standard inner product, so the integrals in W simplify to:

$$\begin{aligned} \int_0^1 B_1^2(s) ds &= \sum_{k=1}^{\infty} \frac{1}{k^2 \pi^2} \xi_k^2 \\ \int_0^1 B_1(s) ds &= \sum_{k=1}^{\infty} \xi_k \underbrace{\int_0^1 \frac{\sqrt{2}}{k\pi} \sin(k\pi s) ds}_{(\gamma_k)} \\ \gamma_k &:= \frac{\sqrt{2}}{k^2 \pi^2} (1 - (-1)^k) = \begin{cases} \frac{2\sqrt{2}}{k^2 \pi^2} & k \text{ odd} \\ 0 & k \text{ even} \end{cases} \end{aligned}$$

Now that W is a difference of sums, we can rewrite it as a quadratic form:

$$\begin{aligned} W &= \Xi^T (D - \Gamma \Gamma^T) \Xi \\ D &:= \text{diag} \left(\frac{1}{k^2 \pi^2} \right) = \text{diag}((d_k)_k), d_k := \frac{1}{k^2 \pi^2} \\ \Gamma &:= (\gamma_k)_k \\ \Xi &:= (\xi_k)_k \end{aligned}$$

This is in particular, a Gaussian quadratic form of standard Gaussians, whose characteristic function can be found in Schwartz, Bennett, and Stein's "Communication Systems and Techniques":

$$\begin{aligned} \mathbb{E} \left[e^{-\frac{\lambda^2}{2} \Xi^T (D - \Gamma \Gamma^T) \Xi} \right] &= \left(\sqrt{\det(I + \lambda^2 (D - \Gamma \Gamma^T))} \right)^{-1} \\ &= \left(\sqrt{\det(I + \lambda^2 (D - \Gamma \Gamma^T)_{\text{even}}) \det(I + \lambda^2 (D - \Gamma \Gamma^T)_{\text{odd}})} \right)^{-1} \end{aligned}$$

where the second equation is obtained by re-ordering the indices of the matrix to group odd indices together at one end, and even indices at the other, leading to block diagonal structure of $(D - \Gamma \Gamma^T)$ (because for even indices, $\Gamma_i = 0$). For $(D - \Gamma \Gamma^T)_{\text{even}} = D$,

$$\begin{aligned} \det(I + \lambda^2 (D - \Gamma \Gamma^T)_{\text{even}}) &= \prod_k \left(1 + \frac{\lambda^2}{4k^2 \pi^2} \right) \\ &= \frac{\sinh(\lambda/2)}{\lambda/2} \end{aligned}$$

by Euler's sinh product expansion. For $(D - \Gamma \Gamma^T)_{\text{odd}}$, apply the matrix determinant lemma:

$$\begin{aligned} \det(I + \lambda^2 (D_{\text{odd}} - \Gamma_{\text{odd}} \Gamma_{\text{odd}}^T)) &= \underbrace{\det(I + \lambda^2 D_{\text{odd}})}_{(*)} \underbrace{(1 - \lambda^2 \Gamma_{\text{odd}}^T (I + \lambda^2 D_{\text{odd}})^{-1} \Gamma_{\text{odd}})}_{(**)} \\ (*) &= \prod_k \left(1 + \frac{\lambda^2}{(2k+1)^2 \pi^2} \right) = \cosh \left(\frac{\lambda}{2} \right) \\ (**) &= 1 - \lambda^2 \sum_{k=0}^{\infty} \frac{\gamma_{2k+1}^2}{1 + \lambda^2 d_{2k+1}} \\ &= 1 - \lambda^2 \sum_{k=0}^{\infty} \frac{\frac{8}{(2k+1)^4 \pi^4}}{1 + \frac{\lambda^2}{(2k+1)^2 \pi^2}} \\ &= 1 - 8 \sum_{k=0}^{\infty} \left(\frac{1}{(2k+1)^2 \pi^2} - \frac{1}{(2k+1)^2 \pi^2 + \lambda^2} \right) \\ &= 1 - 8 \underbrace{\frac{\pi^2}{8\pi^2}}_{\text{Basel sum}} + \underbrace{\frac{2}{\lambda} \tanh\left(\frac{\lambda}{2}\right)}_{\text{partial fraction expansion of tanh}} \end{aligned}$$

Putting all the terms together, we get

$$\begin{aligned}\mathbb{E} \left[e^{-\frac{\lambda^2}{2} \Xi^T (D - \Gamma \Gamma^T) \Xi} \right] &= \left(\sqrt{\frac{\sinh(\lambda/2)}{\lambda/2} \cosh\left(\frac{\lambda}{2}\right) \frac{2}{\lambda} \tanh\left(\frac{\lambda}{2}\right)} \right)^{-1} \\ &= \frac{\lambda/2}{\sinh(\lambda/2)}\end{aligned}$$

which is the desired characteristic function, and in agreement with Levy's result in "Wiener's Random Function, and other Laplacian random functions" (1950) of Levy Area when conditioned on $(B_1(1), B_2(1)) = (0, 0)$.

4 Prove pathwise uniqueness for solutions of

$$dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt, X_0 = x_0$$

with the weakened condition of σ being α -Holder, $\frac{1}{2} < \alpha < 1$ in x (second argument), and b Lipschitz.

We begin the proof by considering two solutions, X_t^1, X_t^2 of the same SDE, and seek to show $\mathbb{E}[|X_t^1 - X_t^2|]$ vanishes. To approximate $|\cdot|$, we appeal to the family of functions $\phi_\epsilon(x) = (\epsilon + x^2)^{\frac{1}{2}}$, which uniformly converges to $|x|$.

Applying Ito's lemma to $\phi_\epsilon(X_t^1 - X_t^2)$, we get:

$$\begin{aligned}d\phi_\epsilon(X_t^1 - X_t^2) &= \phi'_\epsilon(X_t^1 - X_t^2)d(X_t^1 - X_t^2) + \frac{1}{2}\phi''_\epsilon(X_t^1 - X_t^2)d \underbrace{QV(X_t^1 - X_t^2)}_{\text{Quadratic Variation}} \\ \phi'_\epsilon(x) &= \frac{x}{\sqrt{\epsilon + x^2}} \\ \phi''_\epsilon(x) &= \frac{\epsilon}{(\epsilon + x^2)^{\frac{3}{2}}}\end{aligned}$$

where the quadratic variation is given by

$$\begin{aligned}dQV(X_t^1 - X_t^2) &= \underbrace{dQV(X_t^1)}_{\sigma(t, X_t^1)dt} + \underbrace{dQV(X_t^2)}_{\sigma(t, X_t^2)dt} - 2d \underbrace{CV(X_t^1, X_t^2)}_{\text{Cross Variation}} \\ dCV(X_t^1, X_t^2) &= \sigma(t, X_t^1)\sigma(t, X_t^2)dt\end{aligned}$$

where dCV becomes the product as we consider the same Brownian differential along the path of both solutions. We can thus rewrite $dQV(X_t^1 - X_t^2) = (\sigma(t, X_t^1) - \sigma(t, X_t^2))^2$.

Putting the Ito differential equation in integral form,

$$\begin{aligned}\phi_\epsilon(X_t^1 - X_t^2) &= \phi_\epsilon(X_0^1 - X_0^2) + \int_0^t \frac{(X_s^1 - X_s^2)}{\sqrt{\epsilon + (X_s^1 - X_s^2)^2}} (b(s, X_s^1) - b(s, X_s^2))ds \\ &\quad + \int_0^t \frac{(X_s^1 - X_s^2)}{\sqrt{\epsilon + (X_s^1 - X_s^2)^2}} (\sigma(s, X_s^1) - \sigma(s, X_s^2))dB_s \\ &\quad + \frac{1}{2} \int_0^t \frac{\epsilon}{(\epsilon + (X_s^1 - X_s^2)^2)^{\frac{3}{2}}} (\sigma(s, X_s^1) - \sigma(s, X_s^2))^2 ds\end{aligned}$$

Define $\Delta_t := X_t^1 - X_t^2$ for convenience, and taking expectation of the above, we get

$$\begin{aligned}\mathbb{E}[\phi_\epsilon(\Delta_t)] &= \mathbb{E}[\phi_\epsilon(\Delta_0)] + \underbrace{\mathbb{E}\left[\int_0^t \frac{\Delta_s}{\sqrt{\epsilon + \Delta_s^2}} (b(s, X_s^1) - b(s, X_s^2)) ds\right]}_{(I)} \\ &\quad + \underbrace{\mathbb{E}\left[\int_0^t \frac{\Delta_s}{\sqrt{\epsilon + \Delta_s^2}} (\sigma(s, X_s^1) - \sigma(s, X_s^2)) dB_s\right]}_{(II)} \\ &\quad + \underbrace{\frac{1}{2} \mathbb{E}\left[\int_0^t \frac{\epsilon(\sigma(s, X_s^1) - \sigma(s, X_s^2))^2}{(\epsilon + \Delta_s^2)^{\frac{3}{2}}} ds\right]}_{(III)}\end{aligned}$$

We can justify (II) vanishing over expectation, because $\frac{\Delta_s}{\sqrt{\epsilon + \Delta_s^2}} \leq 1$, and $|\sigma(s, X_s^1) - \sigma(s, X_s^2)| \leq C|\Delta_s|^\alpha$, so the integrand is L^2 for $\alpha \leq 1$ if $|\Delta_s| \in L^2$, and thus we are taking expectation over an Ito integral which is a martingale. Now, to bound the other two terms,

$$\begin{aligned}(I) &\leq \mathbb{E}\left[\int_0^t C_1 |\Delta_s| ds\right] \\ (III) &\leq \mathbb{E}\left[\int_0^t \frac{C_2 |\Delta_s|^{2\alpha} \epsilon}{(\epsilon + \Delta_s^2)^{\frac{3}{2}}} ds\right]\end{aligned}$$

where the first inequality comes straight from Lipschitz condition on b , and the second inequality from Holder on σ . It would be very convenient to get (III)'s bound in terms of a linear $|\Delta_s|$ on the right hand side (like (I)'s bound) or have it vanish. Since we already have ϵ in the numerator, we should examine the convergence of the integrand $\frac{\epsilon |\Delta_s|^{2\alpha}}{(\epsilon + \Delta_s^2)^{\frac{3}{2}}}$:

First, the case when $\Delta_s^2 \geq \epsilon$:

$$(\epsilon + \Delta_s^2)^{\frac{3}{2}} \geq |\Delta_s|^3 \implies \frac{\epsilon |\Delta_s|^{2\alpha}}{(\epsilon + \Delta_s^2)^{\frac{3}{2}}} \leq \epsilon |\Delta_s|^{2\alpha-3} \leq \epsilon^{\alpha-\frac{1}{2}}$$

with the last inequality arising because $\alpha \leq 1$.

In the case where $\Delta_s^2 < \epsilon$:

$$(\epsilon + \Delta_s^2)^{\frac{3}{2}} \geq \epsilon^{\frac{3}{2}} \implies \frac{\epsilon |\Delta_s|^{2\alpha}}{(\epsilon + \Delta_s^2)^{\frac{3}{2}}} \leq \epsilon^{-\frac{1}{2}} |\Delta_s|^{2\alpha} < \epsilon^{\alpha-\frac{1}{2}}$$

In both cases, in the regime $\frac{1}{2} < \alpha < 1$, the integrand of (III) is bounded by $C\epsilon^{\alpha-\frac{1}{2}}$. Thus,

$$\mathbb{E}[\phi_\epsilon(\Delta_t)] \leq \mathbb{E}[\phi_\epsilon(\Delta_0)] + C_1 \int_0^t \mathbb{E}[|\Delta_s|] ds + C_3 \epsilon^{\alpha-\frac{1}{2}} t$$

Taking the limit as $\epsilon \rightarrow 0$ by uniform convergence on the ϕ_ϵ functions yields us

$$\mathbb{E}[|\Delta_t|] \leq \mathbb{E}[|\Delta_0|] + C_1 \int_0^t \mathbb{E}[|\Delta_s|] ds$$

By Gronwall's inequality, for some constant C_4 ,

$$\mathbb{E}[|\Delta_t|] \leq C_4 e^{C_1 t} \mathbb{E}[|\Delta_0|] \leq 0$$

So $\mathbb{E}[|X_t^1 - X_t^2|] = 0 \forall t$, which implies by continuity of solutions that $X_t^1 = X_t^2$ almost surely on an interval $[0, T]$. $\square$

5 Derive the moments $\mathbb{E}[X_t^p]$, for $p \in \mathbb{N}$, of the solution to the SDE

$$dX_t = \sqrt{X_t} dB_t, X_0 = x_0 > 0$$

Begin by considering Ito's formula applied to $f(x) = x^p$:

$$dX_t^p = pX_t^{p-1}dX_t + \frac{1}{2}p(p-1)X_t^{p-2}dQV(X_t)$$

Recall that the quadratic variance of an Ito process is calculated using only its diffusion coefficient, $QV(X_t) = \int_0^t \sigma_s^2 ds$, so $dQV(X_t) = X_t dt$ in our case. Then, by substituting and taking expectation:

$$\begin{aligned} dX_t^p &= pX_t^{p-1}dB_t + \frac{1}{2}p(p-1)X_t^{p-1}dt \\ X_t^p &= x_0^p + \frac{1}{2}p(p-1) \int_0^t X_s^{p-1}ds + p \int_0^t X_s^{p-1}dB_s \\ \mathbb{E}[X_t^p] &= x_0^p + \frac{1}{2}p(p-1) \int_0^t \mathbb{E}[X_s^{p-1}]ds + p \underbrace{\mathbb{E} \left[\int_0^t X_s^{p-1}dB_s \right]}_{(I)} \end{aligned}$$

where (I) vanishes under expectation assuming the integrand is L^2 /the expectation exists, because it is a martingale. Observe that we can build a system of linear ODEs through the relation

$$\frac{d\mathbb{E}[X_t^p]}{dt} = \frac{1}{2}p(p-1)\mathbb{E}[X_t^{p-1}]$$

Thus, let us define the system

$$\begin{aligned} g_p(t) &:= \mathbb{E}[X_t^p] \\ g_p'(t) &= \frac{1}{2}p(p-1)g_{p-1}(t) \\ g_p(0) &= x_0^p \end{aligned}$$

Starting from $\mathbb{E}[X_t] = g_1(t) = x_0$,

$$\begin{aligned} g_2'(t) &= g_1(t) = x_0 \implies g_2(t) = x_0^2 + x_0 t \\ g_3'(t) &= 3(x_0^2 + x_0 t) \implies g_3(t) = x_0^3 + 3x_0^2 t + \frac{3}{2}x_0 t^2 \end{aligned}$$

So the general form has

$$\begin{aligned} g_p(t) &= \sum_{k=0}^{p-1} c_{p,k} x_0^{p-k} t^k \\ g_p'(t) &= \sum_{k=0}^{p-1} k c_{p,k} x_0^{p-k} t^{k-1} \end{aligned} \quad (**)$$

But we also know that

$$g_p'(t) = \frac{1}{2}p(p-1)g_{p-1}(t) = \sum_{j=0}^{p-2} \frac{1}{2}p(p-1)c_{p-1,j}x_0^{p-1-j}t^j \quad (***)$$

We need to quickly deal with the summation index mismatch between k, j -in (**), the exponents t^{k-1} for $k = 0, 1, \dots, p-1$ are $t^0, \dots, t^{p-2}$, and thus, in (***), t^j for $j = 0, \dots, p-2$ requires an index shift $j \mapsto k-1$. Matching the per-term coefficients, we have

$$k c_{p,k} x_0^{p-k} = \frac{1}{2}p(p-1)c_{p-1,k-1}x_0^{p-1-k+1} \implies c_{p,k} = \frac{p(p-1)}{2k}c_{p-1,k-1}, \quad c_{p,0} = 1 \quad \forall p$$

We can thus construct an explicit formula for $c_{p,k}$:

$$\begin{aligned}
c_{p,k} &= \frac{p(p-1)}{2k} \cdot \frac{(p-1)(p-2)}{2k-1} \cdot \dots \cdot \frac{(p-k+1)(p-k)}{2} \\
&= \frac{1}{2^k k!} p(p-1)^2 \dots (p-k+1)^2 (p-k) \\
&= \frac{1}{2^k k!} \frac{p!}{(p-k)!} \frac{(p-1)!}{(p-k-1)!} \\
&= \frac{1}{2^k} \binom{p}{k} \frac{(p-1)!}{(p-k-1)!}
\end{aligned}$$

and finally, plugging it into $g_p(t)$,

$$\mathbb{E}[X_t^p] = \sum_{k=0}^{p-1} x_0^{p-k} \left(\frac{t}{2}\right)^k \binom{p}{k} \frac{(p-1)!}{(p-k-1)!}$$

as desired.

6 Suppose that $X_t \in \mathbb{R}^d$ solves the SDE

$$dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt$$

where $\sigma(t, X_t)$ is a Lipschitz matrix in the second argument for an m -dimensional Brownian motion, and $b(t, X_t)$ is a Lipschitz vector in the second argument. Let A be a closed set in $\mathbb{R}^d$, and define the hitting time

$$\tau = \inf\{t \geq 0, X_t \in A\}$$

Let $f(X_t)$ be a function of only X_t , and $\mathcal{L}$ be the infinitesimal generator. Suppose that $\mathcal{L}f = \lambda f$ for $x \in A^C$, $\lambda > 0$. Prove the inequality

$$\frac{f(x_0)}{\sup_{y \in A} f(y)} \leq \mathbb{E}[e^{-\lambda\tau} | X_0 = x_0] \leq \frac{f(x_0)}{\inf_{y \in A} f(y)}$$

The intuition here is to find something that when Ito's lemma is applied to it, will make use of the eigen-identity and cancel out the generator $\mathcal{L}f$. Consider applying Ito's lemma to $e^{-\lambda t} f(X_t)$, for $t < \tau$:

$$\begin{aligned}
d(e^{-\lambda t} f(X_t)) &= e^{-\lambda t} df(X_t) + f(X_t) d(e^{-\lambda t}) + \underbrace{(df(X_t))(d(e^{-\lambda t}))}_{d(e^{-\lambda t}) \text{ contributes a } dt \text{ term}} \\
&= e^{-\lambda t} (\partial_t + \mathcal{L})f(X_t)dt - \lambda e^{-\lambda t} f(X_t)dt + e^{-\lambda t} \nabla_x f(X_t)^T \sigma(t, X_t)dB_t
\end{aligned}$$

Notice that the last term in the first line vanishes because a dt multiplier results in higher order differentials that all vanish. Now, applying $\mathcal{L}f = \lambda f$, and realizing that $\partial_t f(X_t) = 0$,

$$d(e^{-\lambda t} f(X_t)) = e^{-\lambda t} \nabla_x f(X_t)^T \sigma(t, X_t)dB_t$$

It is convenient to note now that the process $M_t := e^{-\lambda(t \wedge \tau)} f(X_{t \wedge \tau})$ has no drift so it is a martingale. Thus, by the optional stopping theorem $\mathbb{E}[M_\tau | x_0] = \mathbb{E}[M_0 | x_0] = f(x_0)$. Letting $t \rightarrow \infty$, $(t \wedge \tau) \rightarrow \tau$ almost surely, so

$$f(x_0) = \mathbb{E}[e^{-\lambda\tau} f(X_\tau) | x_0]$$

By definition of τ , $X_\tau \in A$, so

$$\inf_{y \in A} f(y) \leq f(X_\tau) \leq \sup_{y \in A} f(y) \iff \mathbb{E}[e^{-\lambda\tau} | x_0] \inf_y f(y) \leq \underbrace{\mathbb{E}[e^{-\lambda\tau} f(X_\tau) | x_0]}_{f(x_0)} \leq \mathbb{E}[e^{-\lambda\tau} | x_0] \sup_y f(y)$$

Rearranging gives us the desired inequality: $\frac{f(x_0)}{\sup_{y \in A} f(y)} \leq \mathbb{E}[e^{-\lambda\tau} | X_0 = x_0] \leq \frac{f(x_0)}{\inf_{y \in A} f(y)}$. $\square$

7 The Stratonovich integral is defined as:

$$\int_0^t \sigma(s, B_s) \circ dB_s := \lim_{|\Pi| \downarrow 0} \sum_{i=0}^n \frac{1}{2} [\sigma(t_i, B_{t_i}) + \sigma(t_{i+1}, B_{t_{i+1}})] (B_{t_{i+1}} - B_{t_i})$$

1. Find the transformation from the Stratonovich integral to the Ito integral, and the sufficient condition on σ
2. Suppose that

$$dX_t = b(t, X_t)dt + \sigma(t, X_t) \circ dB_t$$

Find $\tilde{\sigma}, \tilde{b}$ so that

$$dX_t = \tilde{b}(t, X_t)dt + \tilde{\sigma}(t, X_t)dB_t$$

and the sufficient conditions on σ, b for this transformation

3. Find the Radon-Nikodym derivative between the measures induced by the Stratonovich SDE and its corresponding Ito SDE,

$$\begin{aligned} dX_t &= b(t, X_t)dt + \sigma(t, X_t) \circ dB_t \\ dX_t &= b(t, X_t)dt + \sigma(t, X_t)dB_t \end{aligned}$$

and a sufficient condition for absolute continuity on bounded time intervals

Part 1: Start by rewriting the integral as

$$\begin{aligned} \int_0^t \sigma(s, B_s) \circ dB_s &= \lim_{|\Pi| \downarrow 0} \sum_{i=0}^n \frac{1}{2} [\sigma(t_i, B_{t_i}) + \sigma(t_{i+1}, B_{t_{i+1}})] (B_{t_{i+1}} - B_{t_i}) \\ &= \lim_{|\Pi| \downarrow 0} \sum_{i=0}^n (B_{t_{i+1}} - B_{t_i}) \frac{1}{2} [\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})] + \sum_{i=0}^n \sigma(t_i, B_{t_i}) (B_{t_{i+1}} - B_{t_i}) \end{aligned}$$

and notice that the second term $\rightarrow \int_0^t \sigma dB_s$ in Ito sense. The first term needs a bit more work:

$$\begin{aligned} &\lim_{|\Pi| \downarrow 0} \sum_{i=0}^n \frac{1}{2} (B_{t_{i+1}} - B_{t_i}) [\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})] \\ &= \lim_{|\Pi| \downarrow 0} \sum_{i=0}^n \frac{1}{2} (B_{t_{i+1}} - B_{t_i})^2 \underbrace{\left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right)}_{\text{secant}} \\ &= \lim_{|\Pi| \downarrow 0} \sum_{i=0}^n \frac{1}{2} (B_{t_{i+1}} - B_{t_i})^2 \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right) \\ &= \lim_{|\Pi| \downarrow 0} \sum_{i=0}^n \frac{1}{2} (B_{t_{i+1}} - B_{t_i})^2 \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_{i+1}, B_{t_i}) + \sigma(t_{i+1}, B_{t_i}) - \sigma(t_i, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right) \\ &= \lim_{|\Pi| \downarrow 0} \underbrace{\sum_{i=0}^n \frac{1}{2} (B_{t_{i+1}} - B_{t_i}) (\sigma(t_{i+1}, B_{t_i}) - \sigma(t_i, B_{t_i}))}_{(I)} + \underbrace{\sum_{i=0}^n \frac{1}{2} (B_{t_{i+1}} - B_{t_i})^2 \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_{i+1}, B_{t_i})}{B_{t_{i+1}} - B_{t_i}} \right)}_{(II)} \end{aligned}$$

Applying mean value theorem in t ,

$$(I) = \frac{1}{2} \sum_{i=0}^n \sigma_t(\gamma_i, B_{t_i}) (B_{t_{i+1}} - B_{t_i}) (t_{i+1} - t_i) \rightarrow \int_0^t \sigma_s(s, B_s) ds dB_s = 0$$

for some $\gamma_i \in (t_i, t_{i+1})$, with convergence as $n \rightarrow \infty$.

To be more precise, $\sigma_t(\gamma_i, B_{t_i})$ is adapted to the standard Brownian/Ito filtration, and because Brownian paths are uniformly continuous on compact sets,

$$\sum_j (B_{t_{j+1}} - B_{t_j})(t_{j+1} - t_j) \leq \max_j |B_{t_{j+1}} - B_{t_j}| \sum_j (t_{j+1} - t_j) \rightarrow 0$$

as the mesh size $|\Pi| \downarrow 0$ (also the same reason why $dt dB_t$ higher order differentials vanish).

Note that implicitly here, we are assuming $\sigma \in C_t^1$ through mean value theorem and the continuity of the integrand $\gamma_i \mapsto t$. (Also note that we could have handled (I) by citing continuity of σ in t and treating it as the limit of difference of two $\int_0^t \sigma dB_s$ integrals, which requires less regularity on σ). To deal with (II), define $\Delta B_{t_i} := B_{t_{i+1}} - B_{t_i}$, and

$$\begin{aligned} (II) &= \frac{1}{2} \sum_i \left(\frac{\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_{i+1}, B_{t_i})}{\Delta B_{t_i}} \right) (\Delta B_{t_i})^2 \\ &= \frac{1}{2} \sum_i \sigma_x(t_{i+1}, \xi_i) (\Delta B_{t_i})^2 \\ &= \frac{1}{2} \left(\sum_i \sigma_x(t_{i+1}, \xi_i) (t_{i+1} - t_i) + \sum_i \sigma_x(t_{i+1}, \xi_i) ((\Delta B_{t_i})^2 - (t_{i+1} - t_i)) \right) \end{aligned}$$

again justified by using mean value theorem on σ in its second argument, for some $\xi_i \in (B_{t_i}, B_{t_{i+1}})$. Assuming the condition $\sigma \in C_x^1$ on a compact set implies that it is bounded, and recall that the variance of $(\Delta B_i)^2$ is equal to $t_{i+1} - t_i$ from the derivation of Brownian motion's quadratic variance. As $n \rightarrow \infty$, we have

$$\begin{aligned} \sum_i \sigma_x(t_{i+1}, \xi_i) ((\Delta B_{t_i})^2 - (t_{i+1} - t_i)) &\leq \sup_i \sigma_x(t_{i+1}, \xi_i) \sum_i ((\Delta B_{t_i})^2 - (t_{i+1} - t_i)) \rightarrow \underbrace{0}_{\text{expectation}} \\ \frac{1}{2} \sum_i \sigma_x(t_{i+1}, \xi_i) (t_{i+1} - t_i) &\rightarrow \frac{1}{2} \int_0^t \sigma_x(s, B_s) ds \end{aligned}$$

where again, assuming continuity in the second argument makes $\xi_i \mapsto B_{t_i}$ justifiable as we integrate ds . Thus, the Stratonovich integral converges to:

$$\int_0^t \sigma(s, B_s) dB_s + \frac{1}{2} \int_0^t \sigma_x(s, B_s) ds$$

with sufficient assumption being $\sigma \in C_{t,x}^{1,1}$. But can we do better?

Recall the initial sum that we split up:

$$\lim_{|\Pi| \downarrow 0} \sum_{i=0}^n (B_{t_{i+1}} - B_{t_i}) \frac{1}{2} [\sigma(t_{i+1}, B_{t_{i+1}}) - \sigma(t_i, B_{t_i})] + \sum_{i=0}^n \sigma(t_i, B_{t_i}) (B_{t_{i+1}} - B_{t_i}) \rightarrow \int_0^t \sigma(s, B_s) dB_s + \frac{1}{2} CV(\sigma, B_t)$$

where $CV(\sigma, B_t)$ is the cross variation between σ, B_t . To find this covariation directly, examine σ in Ito differential form:

$$d\sigma(t, B_t) = \sigma_x(t, B_t) dB_t + \left(\sigma_t(t, B_t) + \frac{1}{2} \sigma_{tt}(t, B_t) \right) dt$$

Because cross variation is essentially a $d\sigma(t, X_t) dB_t$ cross term in integral form, its contribution will be only coming from the diffusion terms. In other words,

$$d\sigma(t, B_t) dB_t = \sigma_x(t, B_t) (dB_t)^2 + \left(\sigma_t(t, B_t) + \frac{1}{2} \sigma_{tt}(t, B_t) \right) (dt)(dB_t)$$

and it is obvious that cross terms ($dt dB_t$) vanish, and that $(dB_t)^2 = dt$. Thus,

$$CV(\sigma, B_t) = \int_0^t d\sigma(s, B_s) dB_s = \int_0^t \sigma_x(s, B_s) ds$$

and the Stratonovich conversion is

$$\int_0^t \sigma(s, B_s) \circ dB_s = \int_0^t \sigma(s, B_s) dB_s + \frac{1}{2} \int_0^t \sigma_x(s, B_s) ds$$

which matches the result we got before. But for this derivation, the sufficient condition is only that σ, σ_x are "Brownian-integrable" on $[0, t]$, i.e. $\sigma, \sigma_x \in L_{loc}^2$, and that σ_{xx}, σ_t exist for the use of Ito's lemma (this condition is traded for $\sigma \in C_x^1$ from before).

Part 2: Consider the stratonovich SDE in integral form:

$$X_t = X_0 + \int_0^t b(s, X_s) ds + \underbrace{\int_0^t \sigma(s, X_s) \circ dB_s}_{(*)}$$

so the question reduces to—what is the Stratonovich integral $(*)$? Similar to part 1, let us write:

$$\begin{aligned} (*) &= \lim_{|\Pi| \downarrow 0} \sum_{i=0}^{n-1} \left(\frac{\sigma(t_{i+1}, X_{t_{i+1}}) - \sigma(t_i, X_{t_i})}{2} \right) (B_{t_{i+1}} - B_{t_i}) + \sum_{i=0}^{n-1} (B_{t_{i+1}} - B_{t_i}) \sigma(t_i, X_{t_i}) \\ &= \frac{1}{2} CV(\sigma(t, X_t), B_t) + \int_0^t \sigma(s, X_s) dB_s \end{aligned}$$

As before, apply Ito's lemma to $\sigma(t, X_t)$:

$$d\sigma(t, X_t) = \sigma_x(t, X_t) dX_t + \sigma_t(t, X_t) dt + \frac{1}{2} \sigma_{xx}(t, X_t) dQV(X_t)$$

Assuming it exists (reasonable considering the question), the Ito formulation of X_t is $dX_t = \tilde{b}(t, X_t) dt + \tilde{\sigma}(t, X_t) dB_t$, and because X_t is an Ito process, $dQV(X_t) = \tilde{\sigma}^2(t, X_t) dt$. Thus, the differential form of cross variation is

$$\begin{aligned} d\sigma(t, X_t) dB_t &= \sigma_x(t, X_t) (\tilde{b}(t, X_t) dt + \tilde{\sigma}(t, X_t) dB_t) dB_t + \left(\sigma_t(t, X_t) + \frac{1}{2} \sigma_{xx}(t, X_t) \tilde{\sigma}^2(t, X_t) \right) dt dB_t \\ &= \sigma_x(t, X_t) \tilde{\sigma}(t, X_t) dt \end{aligned}$$

as higher order cross terms all vanish. To determine what $\tilde{\sigma}$ is, consider that:

$$\begin{aligned} \int_0^t \sigma(s, X_s) \circ dB_s &= \int_0^t \sigma(s, X_s) dB_s + \frac{1}{2} \int_0^t d\sigma(s, X_s) dB_s \\ &= \int_0^t \sigma(s, X_s) dB_s + \frac{1}{2} \int_0^t \sigma_x(s, X_s) \tilde{\sigma}(s, X_s) ds \end{aligned}$$

and so the differential SDE of X_t is:

$$dX_t = \left(b(t, X_t) + \frac{\sigma_x(t, X_t) \tilde{\sigma}(t, X_t)}{2} \right) dt + \sigma(t, X_t) dB_t$$

which is written purely in Ito form. It only remains to match coefficients, which means $\tilde{\sigma}(t, X_t) = \sigma(t, X_t)$, $\tilde{b}(t, X_t) = b(t, X_t) + \frac{1}{2} \sigma_x(t, X_t) \sigma(t, X_t)$. The sufficient assumptions in this case are that $\sigma_t, \sigma_x, \sigma_{xx}$ should exist for Ito's lemma, and $b, \sigma, \sigma_x \sigma \in L_{loc}^2$ for integration. Furthermore, we also want existence and uniqueness of X_t , so σ, b being 1-Lipschitz or σ being α -Holder is required.

Part 3: Define the path measure induced by an Ito SDE $dX_t = \sigma dB_t + bdt$ to be $\mathbb{P}_x^{\sigma^2, b}$ on $\mathcal{C}([0, T])$. For absolute continuity and existence of the Radon-Nikodym derivative $\frac{d\mathbb{P}_x^{\sigma^2, b}}{d\mathbb{P}_x^{\sigma^2, 0}}$, we appeal to Girsanov's theorem. Since the Stratonovich SDE takes the Ito form $dX_t = \sigma dB_t + (b + \frac{1}{2}\sigma_x \sigma)dt$, its induced measure is $\mathbb{P}_x^{\sigma^2, (b + \frac{1}{2}\sigma_x \sigma)}$. Girsanov's change of measure gives us the R-N derivative:

$$\begin{aligned} \frac{d\mathbb{P}_x^{\sigma^2, (b + \frac{1}{2}\sigma_x \sigma)}}{d\mathbb{P}_x^{\sigma^2, b}} &= e^{-\int_0^t \frac{1}{\sigma} (b + \frac{1}{2}\sigma_x \sigma - b) dB_s - \frac{1}{2} \int_0^t \frac{1}{\sigma^2} (b + \frac{1}{2}\sigma_x \sigma - b)^2 ds} \\ &= e^{-\frac{1}{2} \int_0^t \sigma_x dB_s - \frac{1}{8} \int_0^t \sigma_x^2 ds} \end{aligned}$$

for which a sufficient condition is Novikov's condition (which ensures that the R-N derivative is a martingale), given by:

$$\mathbb{E}[e^{\frac{1}{8} \int_0^t \sigma_x^2 ds}] < \infty$$

The condition also guarantees absolute continuity on bounded time intervals, and all conditions from parts 1 & 2 are required for this to make sense.

8 Let $X_t \in \mathbb{R}^d$ solve the SDE

$$dX_t = \sigma(t, X_t)dB_t + b(t, X_t)dt$$

where $\sigma(t, X_t)$ is a Lipschitz matrix in the second argument for $B_t \in \mathbb{R}^m$, and $b(t, X_t)$ is a Lipschitz vector in the second argument. Suppose $f \in C_{t,x}^{1,2}$. Find a function $g(t, X_t)$, $g(0, X_0) = 1$ where the product $f(t, X_t)g(t, X_t)$ is a martingale.

Recall that an Ito process is a martingale when it is drift-less, i.e. $Y_t = \tilde{\sigma}(t, Y_t)dB_t$, assuming nice enough conditions. We seek to make $Y_t := f(t, X_t)g(t, X_t)$ drift-less. First, apply Ito's formula on $f(t, X_t)$:

$$\begin{aligned} df(t, X_t) &= (\partial_t + \mathcal{L})f(t, X_t)dt + \nabla_x f(t, X_t)^T \sigma(t, X_t)dB_t \\ \mathcal{L} &= \underbrace{\frac{1}{2} \sum_{i,j=1}^d a_{ij} \frac{\partial^2}{\partial x_i \partial x_j}}_{\text{diffusion}} + \sum_{i=1}^d b_i \frac{\partial}{\partial x_i} \\ \mathcal{L}f &= \frac{1}{2} \text{tr}(\sigma \sigma^T \nabla_x^2 f) + b \cdot \nabla_x f \\ a(t, X_t) &= \sigma(t, X_t) \sigma(t, X_t)^T \end{aligned}$$

and we can identify the drift and diffusion coefficients easily from the first line. Define them as:

$$\begin{aligned} A(t, X_t) &:= \partial_t f(t, X_t) + \mathcal{L}f(t, X_t) \\ H(t, X_t) &:= \sigma(t, X_t)^T \nabla_x f(t, X_t) \end{aligned}$$

Our best guess here for the form of $g(t, X_t)$ is that it will be a martingale, and when Ito's lemma is applied to it, it will help cancel the drift in the intermediate result. So we should let

$$dg(t, X_t) = g(t, X_t) \theta(t, X_t)^T dB_t$$

Then, Ito's formula on fg yields (omitting (t, X_t) dependence in notation):

$$\begin{aligned} d(f(t, X_t)g(t, X_t)) &= f dg + g df + (df)(dg) \\ &= fg \theta^T dB_t + g A dt + g H^T dB_t + (A dt + H^T dB_t)(g \theta^T dB_t) \\ &= g A dt + g H^T \theta dt + g H^T dB_t + fg \theta^T dB_t \end{aligned}$$

where the $dt dB_t$ factor vanishes due to it being a higher order stochastic differential. For the martingale property, it is sufficient that

$$g(t, X_t)A(t, X_t) + g(t, X_t)H(t, X_t)^T \theta(t, X_t) = 0 \iff A(t, X_t) + H(t, X_t)^T \theta(t, X_t) = 0$$

which (the latter) is a linear equation in the unknown $\theta(t, X_t)$. Since $H, \theta \in \mathbb{R}^m$ are one-dimensional, we choose the canonical solution $\theta = \lambda H$, so that

$$\lambda(t, X_t) \|H(t, X_t)\|^2 = -A(t, X_t) \implies \theta(t, X_t) = -\frac{A(t, X_t)}{\|H(t, X_t)\|^2} H(t, X_t)$$

and we have that

$$dg(t, X_t) = -g(t, X_t) \frac{A(t, X_t)}{\|H(t, X_t)\|^2} H(t, X_t)^T dB_t, \quad g(0, X_0) = 1$$

To solve for g , the trick is to apply Ito's lemma on $\ln(g(t, X_t))$:

$$\begin{aligned} d \ln(g(t, X_t)) &= \frac{1}{g} dg + \frac{1}{2} \left(-\frac{1}{g^2} \right) (dg)^2 \\ &= -\theta^T dB_t - \frac{1}{2} \frac{(g\theta)^2}{g^2} (dB_t)^2 \\ &= -\theta^T dB_t - \frac{1}{2} \|\theta\|^2 dt \end{aligned}$$

and in integral form,

$$\begin{aligned} \ln(g(t, X_t)) &= \ln(g(0, X_0)) - \int_0^t \theta(s, X_s)^T dB_s - \frac{1}{2} \int_0^t \|\theta(s, X_s)\|^2 ds \\ \implies g(t, X_t) &= e^{-\frac{1}{2} \int_0^t \|\theta(s, X_s)\|^2 ds - \int_0^t \theta(s, X_s)^T dB_s} \\ \theta(t, X_t) &= -\frac{(\partial_t + \mathcal{L})f(t, X_t)}{\|\sigma(t, X_t)^T \nabla_x f(t, X_t)\|^2} \nabla_x f(t, X_t)^T \sigma(t, X_t) \end{aligned}$$

is the function that makes Y_t a martingale.